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Aim:- Write a LEX program to validate mobile number URL, date & time.

Algorithm

1. Start
2. Define regex patterns
3. Read input
4. Match with patterns
5. Display valid/invalid
6. Stop

Program

```
%{
#include<stdio.h>
%}
%%
[0-9]{10}
{ printf("Valid mobile number\n");
}
{http|https://[a-zA-Z0-9./]+;
{
printf("Valid URL\n");
}
[0-9]{2}/[0-9]{2}/[0-9]{4}
{
printf("Valid Date\n");
}
[0-9]{2}:[0-9]{2}
{ printf("Valid Time\n");
}
```

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output

Input : 9988164180

Valid Mobile Number.

Date

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```
. $
    printf("Invalid Input \n");
    }
%%
int main()
{
    yylex();
    return 0;
}
```

Viva Question

Q1 What regular expression do you use to validate an Indian mobile number?

Ans A common regular expression is ^[6-9][0-9]{9}$, It checks that the number contains exactly 10 digits.
The first digit must be between 6 & 9, as required for Indian mobile number.

Q2 How do you validate a URL in LEX?

Ans A URL can be validate using a regular expression that checks for protocols like http, https.
for example https?://[a-zA-Z0-9-]+\.[a-zA-Z]{2,}
It checks the protocol, domain name & domain extension.

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Q3 What are the different date formats you validate?

Ans Common formats include DD-MM-YYYY, DD/MM/YYYY, & YYYY-MM-DD. LEX uses regular expressions to check whether the data follows the required formats. for example, DD-MM-YYYY can be represented as [0-9]{2}-[0-9]{2}-[0-9]{4}.

Q4 How do you handle 12-hour v/s 24 hour time formats?

Ans In the 12-hour format, time is represented using AM or PM, such as 10:30PM.
In the 24-hour format, hour range from 00 to 23 such as 22:30.
Different regular expressions are used in LEX to validate these two formats.

Q5 What is the purpose of start conditions in validating complex patterns?

Ans Start condition in LEX allow specific rules to be activated only in particular states.
They are useful when different patterns needs to be recognized in different contexts.
for example, they can help validate the contents of a string, comment or other complex structure.

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